DATA ANALYST COURSE

WEEK 4: DATA VISUALIZATION & CAPSTONE PROJECT

Complete Sample Solution

Includes: dashboard plan, chart selection, business insights, Python/Pandas analysis, SQL queries, and final report summary.

Note: The dataset link in the provided photo is not readable enough to access. This solution uses a clearly labelled sample retail-sales dataset. Replace it with your course dataset if required.

1. Assignment Overview

Goal: analyze retail sales data, build a dashboard in Power BI or Tableau, and explain useful findings to a business stakeholder.

Sample dataset fields

Column	Meaning	Example
Order_ID	Unique order number	1001
Order_Date	Date of order	2026-01-05
Region	Sales area	East
Category	Product group	Technology
Product	Item sold	Headphones
Units	Quantity sold	3
Unit_Price	Price per item	50
Revenue	Units × Unit_Price	150
Cost	Total product cost	90
Profit	Revenue – Cost	60

Important calculated columns

Revenue = Units × Unit_Price
Profit = Revenue – Cost
Profit Margin % = Profit ÷ Revenue × 100 (when Revenue is not zero).

2. Dashboard Solution (Power BI or Tableau)

Recommended dashboard layout

Top row — KPI cards: Total Revenue, Total Profit, Total Orders, and Profit Margin %. Middle row — line chart for revenue by month and bar chart for revenue by category. Bottom row — bar chart for profit by region and a detailed product table.

Filters / slicers: Order Date, Region, Category, and Product. Keep the same number format across visuals; use currency for revenue and profit, and percentages for margin.

Chart selection and purpose

Visual	Fields	Question answered
KPI cards	Revenue, Profit, Orders, Margin	How is the business performing overall?
Line chart	Month on X-axis; Revenue on Y-axis	How do sales change over time?
Bar chart	Category and Revenue	Which category sells the most?
Bar chart	Region and Profit	Which region contributes most profit?
Table	Product, Units, Revenue, Profit	Which products need attention?

3. Sample Business Insights

The following is an example of the stakeholder summary format. Because the actual course dataset was not accessible from the photo, do not treat these as measured findings from your own data. Calculate the metrics from your dataset and replace the sample statements.

Example insights (illustrative)

1. Revenue is increasing over the period: check the monthly line chart and identify the strongest and weakest months.
2. One product category may generate the highest revenue, but revenue alone does not show profitability. Compare profit and profit margin before recommending more investment.
3. Regional performance can differ: focus on regions with high revenue but low profit to investigate discounts or costs.
4. A small number of products may account for a large share of sales. Use a product table sorted by revenue and profit to identify best sellers and low-margin items.

Stakeholder summary template

“The dashboard reviews sales performance by month, category, region, and product. The key metrics are total revenue, total profit, order count, and profit margin. The strongest category/region was [insert result], while [insert result] needs improvement. I recommend promoting profitable products, reviewing costs and discounts in low-margin areas, and monitoring monthly sales to plan inventory.”

4. Python / Pandas: Clean and Explore the Dataset

Install the required packages (once): **pip install pandas matplotlib seaborn**. Save your dataset as **sales.csv** in the same folder as the Python file.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Read the CSV file
df = pd.read_csv("sales.csv")

# 2. Inspect the data
print(df.head())
print(df.info())
print(df.isnull().sum())

# 3. Remove duplicate rows
df = df.drop_duplicates()

# 4. Clean column names
df.columns = df.columns.str.strip().str.replace(" ", "_")

# 5. Convert dates and numeric columns
df["Order_Date"] = pd.to_datetime(df["Order_Date"], errors="coerce")
for col in ["Units", "Unit_Price", "Cost"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

# 6. Handle missing values
df = df.dropna(subset=["Order_Date", "Region", "Category", "Units",
                       "Unit_Price", "Cost"])
df = df[(df["Units"] >= 0) & (df["Unit_Price"] >= 0) & (df["Cost"] >= 0)]

# 7. Create calculated fields
df["Revenue"] = df["Units"] * df["Unit_Price"]
df["Profit"] = df["Revenue"] - df["Cost"]
df["Month"] = df["Order_Date"].dt.to_period("M").astype(str)
df["Profit_Margin_Pct"] = (
    df["Profit"] / df["Revenue"].replace(0, pd.NA) * 100
)

# 8. Summary tables
print("Total revenue:", df["Revenue"].sum())
print("Total profit:", df["Profit"].sum())
print("Order count:", df["Order_ID"].nunique())
print(df.groupby("Category")["Revenue"].sum().sort_values(ascending=False))
print(df.groupby("Region")["Profit"].sum().sort_values(ascending=False))

# 9. Visualizations
monthly = df.groupby("Month", as_index=False)["Revenue"].sum()
```

```
sns.lineplot(data=monthly, x="Month", y="Revenue", marker="o")
plt.title("Monthly Revenue")
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig("monthly_revenue.png")
plt.close()

category = df.groupby("Category", as_index=False)["Revenue"].sum()
sns.barplot(data=category, x="Category", y="Revenue")
plt.title("Revenue by Category")
plt.xticks(rotation=30)
plt.tight_layout()
plt.savefig("revenue_by_category.png")
plt.close()

# 10. Export cleaned data for Power BI / Tableau
df.to_csv("cleaned_sales.csv", index=False)
print("Saved cleaned_sales.csv and chart images.")
```

What this code does

It loads the CSV, checks missing values, removes duplicates, converts data types, removes unusable records, calculates revenue and profit, summarizes performance, creates two charts, and exports a cleaned CSV for the dashboard.

5. SQL Analysis Queries

These queries assume a table named sales with columns Order_ID, Order_Date, Region, Category, Units, Unit_Price, and Cost. SQL syntax can vary slightly by database.

```
-- Total revenue and profit
SELECT
  SUM(Units * Unit_Price) AS total_revenue,
  SUM((Units * Unit_Price) - Cost) AS total_profit,
  COUNT(DISTINCT Order_ID) AS total_orders
FROM sales;

-- Revenue by category
SELECT Category,
       SUM(Units * Unit_Price) AS revenue
FROM sales
GROUP BY Category
ORDER BY revenue DESC;

-- Profit by region
SELECT Region,
       SUM((Units * Unit_Price) - Cost) AS profit
FROM sales
GROUP BY Region
ORDER BY profit DESC;

-- Monthly revenue (syntax may vary by database)
SELECT
  EXTRACT(YEAR FROM Order_Date) AS sales_year,
  EXTRACT(MONTH FROM Order_Date) AS sales_month,
  SUM(Units * Unit_Price) AS revenue
FROM sales
GROUP BY EXTRACT(YEAR FROM Order_Date),
         EXTRACT(MONTH FROM Order_Date)
ORDER BY sales_year, sales_month;
```

6. Power BI Measures (DAX)

Create these measures after importing the cleaned dataset. Replace **Sales** with your actual table name if different.

```
Total Revenue = SUMX(Sales, Sales[Units] * Sales[Unit_Price])

Total Profit =
SUMX(Sales, (Sales[Units] * Sales[Unit_Price]) - Sales[Cost])

Total Orders = DISTINCTCOUNT(Sales[Order_ID])

Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)
```

Dashboard build steps

1. Open Power BI Desktop and select Get Data → Text/CSV.
2. Import cleaned_sales.csv and confirm data types.
3. In Report view, add four Card visuals for the measures above.
4. Add a Line chart (Month, Total Revenue), a Bar chart (Category, Total Revenue), and a Bar chart (Region, Total Profit).
5. Add slicers for Order_Date, Region, and Category.
6. Add clear titles, data labels where useful, and consistent currency formatting.
7. Save the report as a .pbix file and export a PDF if required.

7. Final Capstone Report

Title: Retail Sales Data Analysis and Dashboard

Objective

The objective is to clean and analyze retail sales data, find patterns in revenue and profit, and communicate the results through charts and a dashboard.

Methodology

The dataset is inspected for missing values, duplicate records, incorrect data types, and invalid numeric values. Pandas is used for cleaning and calculations. SQL is used to summarize sales by category, region, and month. Power BI or Tableau is used to create interactive visuals.

Key metrics

Total Revenue = sum of Units × Unit_Price.
Total Profit = sum of Revenue – Cost.
Total Orders = count of unique Order_ID values.
Profit Margin = Total Profit ÷ Total Revenue × 100.

Findings

Insert the actual values from your dataset here: total revenue [■/currency], total profit [■/currency], total orders [number], best-performing category [name], highest-profit region [name], and best/worst month [month]. Support each finding with a chart or summary table.

Recommendations

1. Increase focus on categories with both strong revenue and healthy profit margins.
2. Investigate regions with high sales but weak profit, including discounts and operating costs.
3. Use monthly trends to improve inventory and promotional planning.
4. Review low-performing products and decide whether to improve pricing, marketing, or stock levels.

Conclusion

This project demonstrates the end-to-end data analysis process: data cleaning, exploratory analysis, SQL summaries, visualization, dashboard design, and business recommendations. Final conclusions should be updated after running the analysis on the actual course dataset.

8. Submission Checklist

■ Cleaned dataset (cleaned_sales.csv)
■ Python source file (.py or notebook)
■ Power BI dashboard (.pbix) or Tableau workbook
■ Dashboard screenshots or PDF export
■ Final report with real metrics and recommendations
■ Demo recording or explanation, if requested
■ GitHub repository or ZIP containing the project files

Before submitting: replace every bracketed placeholder with results from your own dataset. Do not submit illustrative insights as if they were calculated results.